

PROJECT MANAGEMENT

Workbook

2025

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This workbook is copyrighted by RWarb Publishing 2014. It goes with the Art & Science of Project Management by Warburton & Kanabar, 2nd Edition. Please email your professor recommended edits if any. It helps students understand PMBOK processes—as students are creating process outputs within the context of an assigned case study. The First Part is in the textbook.

Project Description

The project focuses on the implementation of an **AI-based Bike Riding Detection System** to enhance road safety and ensure compliance with traffic regulations. Many accidents occur due to unauthorized or unsafe bike riding, such as riding without a helmet, riding in restricted areas, or reckless driving. Traditional surveillance methods require manual monitoring, which is time-consuming and inefficient.

Our system utilizes **computer vision and AI-based detection techniques** to automatically identify individuals riding bikes and analyse key parameters such as helmet usage, riding behaviour, and adherence to road safety rules. By integrating real-time image processing with machine learning algorithms, the system can detect violations and generate alerts or reports for further action.

This automated approach significantly reduces the need for manual supervision, improves road safety, and helps enforce traffic laws efficiently. The AI-powered detection system can be integrated with existing surveillance infrastructure, making it a scalable and effective solution for smart city traffic management.

PART B

Initiating Process Group

Introduces you to two PMBOK Processes

- a) Create Charter
- b) Identify Stakeholders

2.1. PROJECT CHARTER

Project Title	AI-POWERED AUTOMATED UNDERAGE BIKE RIDING DETECTION SYSTEM
Organization	SAFE ROADS INITIATIVE
Start Date.	1 January 2025
End Date.	30 September 2025
Project. Champion	Jayadeep
Road safety is a basic necessity for humans and the well-being of society. While there are traffic rules and systems in place, underage bike riding remains a major concern, leading to accidents, injuries, and even fatalities. Identifying underage riders manually is difficult and often inefficient. Ensuring that only eligible riders are on the roads is important for public safety. Underage riders not only endanger their own lives but also the lives of others. Detecting and preventing them from riding illegally requires an efficient, intelligent solution. This system is designed to automatically detect underage bike riders using real-time CCTV footage, facial analysis, and age prediction technologies. Whenever a suspected underage rider is detected, an alert is generated to the authorities. This ensures quick action and helps maintain safer roads. This project provides an efficient and automated way of monitoring traffic and identifying underage riders without manual intervention, thereby preventing accidents and promoting responsible driving behavior.	
High level Requirements	Professional skills on using the machine.

Success Criteria& Who Measures it.	Accurate detection of underage bike riders measured by traffic authorities and system operators through real-time monitoring and incident validation.	
Stakeholder List	Traffic police authorities, municipal officials, AI development team, system operators, and the general public.	
Project Budget	₹40,50,000/-	
Assigned Project Manager, Responsibility and Authority Level	Jayadeep is the assigned project manager. He will report to the CEO.	
Signatures	CEO & Head of operation	
	M Akhil	CH Jayadeep

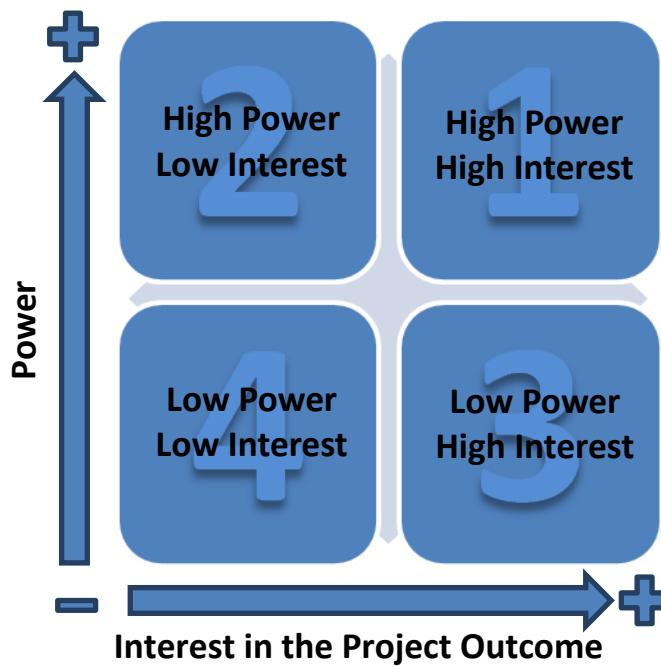
2.2. STAKEHOLDERS

Stakeholder	Role
TRAFFIC POLICE	Major stake holders
M AKHIL	CEO
CH JAYADEEP	Project & Operations manger
M SUMITH	Creative Director & Lead Planner
J VARUN RAJ	Media manager

How to Rank the Stakeholders?

In determining the rank of the stakeholders, use the following approach. Identify stakeholders that have substantial interest in the outcome of the project and have substantial power to help achieve the project goals.

- High Power and High Interest in Project Outcome: CH JAYADEEP
- High Power but Low Interest in Project Outcome: M AKHIL
- Low Power but High Interest in Project Outcome: M SUMITH
- Low Power and Low Interest in Project Outcome: J VARUN RAJ



2.3. STAKEHOLDERS REGISTER & MANAGEMENT STRATEGY

Stakeholder	Stakeholder Rank (High/Med/Low)	Role	Goal (Interest or outcome for the Stakeholder)
Traffic Police	High interest and High power	funding the project	Approves the project scope, provides guidance on planning, anticipates obstacles, and corrects flaws based on their interests.
M Akhil	High power and High interest	CEO	Acts as a project champion, identifies requirements, ensures proper allocation of team members.
CH Jayadeep	Low power and High interest	Project and Operations manager	Approves project scope, oversees innovation and creativity in project execution.
M Sumith	Low power and High interest	Creative Director & Lead Planner	Focuses on appearance and design modifications to improve the overall look of the project.
J Varun Raj	Low Power and Low Interest	Media manager	Ensures the project content is presented attractively to the public, aiming for wider engagement.

Planning Process Group

Collect Requirements Process

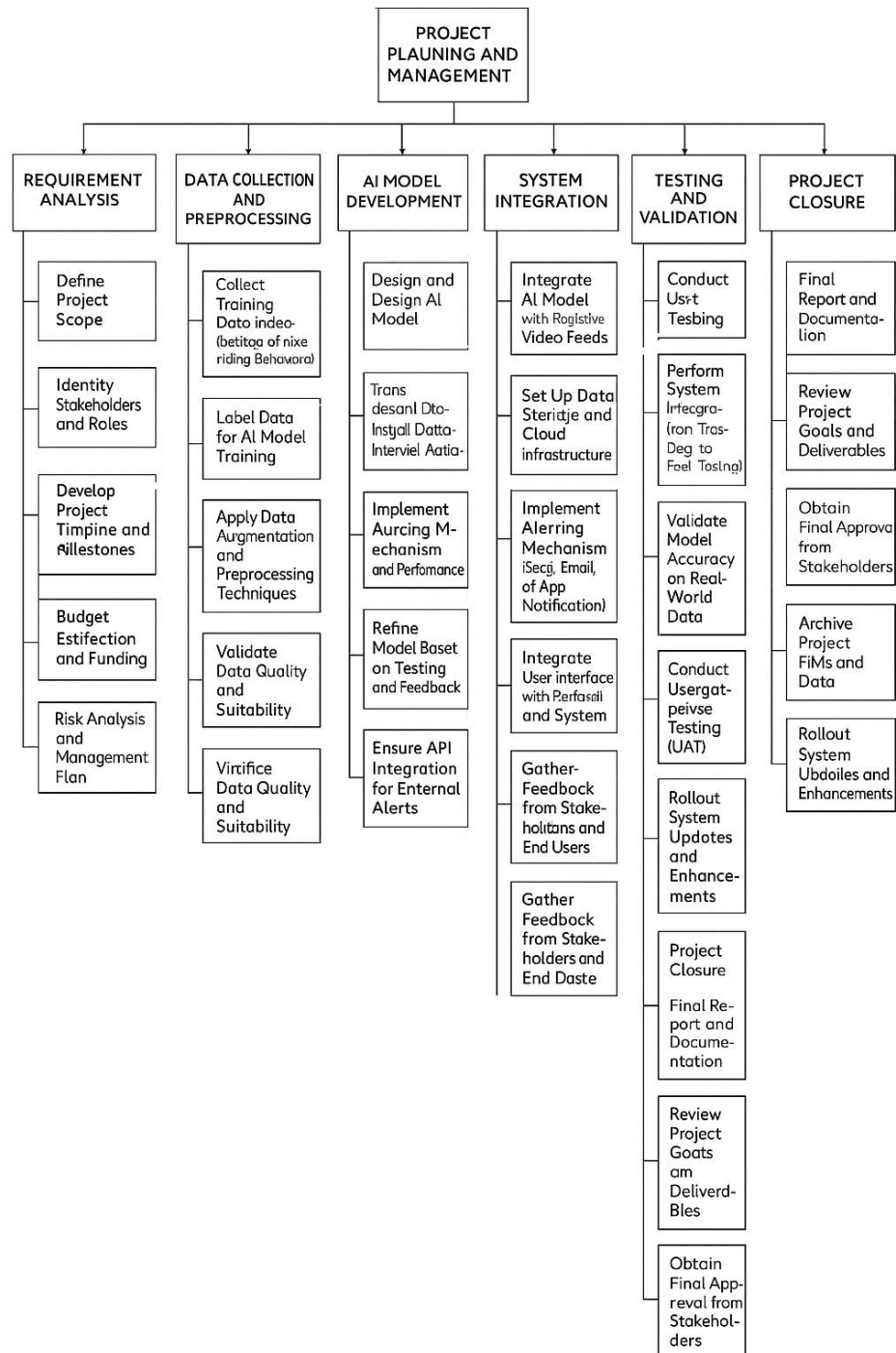
Requirement Document

Category	Requirement	Stakeholder	Acceptance Criteria
Funding	Secure funding for development and implementation of the AI system	Investor	100%
Technical Data	Gathering accurate data for AI model training (video footage, sensor data)	Data Scientist	70%
Location Finalization	Determine deployment locations (e.g., traffic cameras, city areas)	Operations Manager	75%
Execution of Project	Collaboration and support from authorities and road safety personnel	Project Manager	100%
Commissioning	Final system validation and deployment readiness	CEO	50%

2.4. PROJECT SCOPE STATEMENT

PROJECT OBJECTIVE
DELIVERABLES The system will provide an efficient, AI-powered solution for detecting unsafe bike riding behaviors in real-time.
TECHNICAL REQUIREMENTS Deep Learning Models for motion detection Video Cameras for real-time monitoring
LIMITS AND EXCLUSIONS (Constraints) The system is designed to be scalable but may require high computational power for real-time video processing in large cities. The system's effectiveness is dependent on the quality of input video data and real-time processing capabilities.
RISKS accurate Detection: AI may incorrectly flag safe riding behaviors or miss dangerous actions due to environmental factors like weather conditions or low-quality video feeds. Hardware Failure: Camera malfunctions or connectivity issues can lead to gaps in data collection and system performance.
ACCEPTANCE CRITERIA Alerts must be generated in real-time with less than a 3-second delay from detection to notification. The system should have a user-friendly interface for authorities to review detected incidents and take action.
Signatures M Akhil CH Jayadeep

2.5. WORK BREAKDOWN STRUCTURE (WBS)



2.6. DEFINE AND ESTIMATE ACTIVITIES AND RESOURCES

Activity List

Activities	Effort	Resource
Data Collection	High	Video datasets, Cloud storage
Model Training	High	GPUs, Cloud computing
System Integration	Medium	Backend developers, UI/UX designers
Testing	High	Testing environments, Testers
Deployment	Medium	Cloud or Edge devices, Servers
Maintenance	Low	Data storage, Developers
User Training & Feedback	Low	End-users, Documentation

2.7. ACTIVITY DURATIONS

Activity Durations

Activity	Duration (days)
Data Collection	5
Model Training	10
System Integration	10
Testing & Validation	6
Deployment	8
User Training & Feedback	4

2.8. DELIVERABLES LIST BY TEAM

An alternate Way to view the activity list is to view it by resource

Team (Work Resource)	Activities (List Key Deliverables Only)
Team 1	1. Data Collection 2. Model Training
Team 2	3. System Integration 4. Testing & Validation

2.9. PLAN RISK MANAGEMENT

Risk	Type of Risk (Good or Bad (threat))	Consequence or Impact
Final delivery quality out of control	Bad	Losing customer trust
Machine failure or breakdowns	Bad	Replacing is difficult
Maintenance cost is unavoidable	Bad	Affects the cost invested on machine and product quality

Probability 1 ----- 0	High				
	Medium				
	Low				
			Low	Medium	High
	0	Impact -----> 1			

2.10. PLAN QUALITY MANAGEMENT

Plan Quality

Quality Roles and Responsibilities

Role	Responsibility
Project Manager	To oversee and manage all aspects of the AI bike riding detection system project.
AI Model Developer	To design, develop, and optimize the AI models for motion detection and bike riding behavior analysis.
System Integrator	To integrate the AI model with the backend, user interface, and alerting mechanisms.
UI/UX Designer	To design the user interface and ensure a seamless, intuitive experience for end-users.

2.11. QUALITY MANAGEMENT PLAN

Summarizes the previous Section

Quality Management Plan

ProjectTitle:	Version:	Date:
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Overview:

The quality of the AI-based bike riding detection system is crucial to ensure effective and accurate performance. All components of the system, including AI models, integration, and user interfaces, will undergo rigorous quality checks by the project manager and system integrator to guarantee optimal functionality and user experience.

Quality Responsibilities and Roles:

Quality monitoring will be handled by the **Project Manager** and the **AI Model Developer**. These responsibilities will be managed by **Sukesh (Project Manager)** and **[Name of AI Model Developer]**.

Quality Assurance Approach:

Quality will be assured through regular code reviews, model performance validation, and continuous integration testing. We will ensure that all parts of the system meet predefined specifications and standards before deployment.

Quality Control Approach:

Each component of the system, including AI models and integration, will undergo frequent testing and performance evaluations. This includes both manual and automated testing methods to check for system accuracy, speed, and usability.

Quality Reporting Plan:

Quality checks will be conducted every week to evaluate the AI model performance, system integration, and overall user experience. Detailed reports on quality assurance and control will be generated and shared with the project team after every major milestone or update.

2.12. DEVELOP HUMAN RESOURCES PLAN

Human Resources Plan

- Project Manager
- AI Model Developers
- System Integrators
- UI/UX Designers
- Quality Assurance (QA) Testers
- Public Relations Staff

2.13. COMMUNICATION MANAGEMENT PLAN

Communication Management Plan

Message Description	Report To	Method	Frequency
Weekly Progress Update	Project Manager, Stakeholders	Email/Meetings	Weekly
Bug Reports	QA Lead, Developers	Bug Tracker	Daily
System Performance Reports	Stakeholders, Project Manager	Email/Meetings	Bi-weekly
Deployment Updates	Developers, System Integrators	Email/Meetings	As needed
User Feedback Reports	Project Manager, UI/UX Designers	Feedback forms/Surveys	Monthly

2.14. COST ESTIMATION

Category	Estimated Cost
Development	₹15,00,000
Infrastructure & Hosting	₹5,00,000
AI Model Training & Optimization	₹8,00,000
Security & Compliance	₹3,00,000
Marketing & Outreach	₹4,00,000
Operational/Admin	₹3,50,000
Testing & Validation	₹2,00,000
User Training & Documentation	₹1,00,000
Total	₹40,50,000

2.15. PROJECT SUMMARY

Now that we have completed the project plan you can summarize the details such as:

Performance Indicator	Planned
Total Effort	3200 Hours
Total Duration	120 Days
Start Date	1st January 2025
End Date	30th September 2025
Cost	₹40,50,000

2.16. RISK MONITORING AND CONTROL

Updated Risk Register

Risk ID	Risk	Risk Response	Resource Responsible for Mitigation	Current Risk Status
1	Over Budget	To maintain costs within the allocated budget	Project Manager, Finance Team	Within limits
2	Delays in Development	To ensure timely delivery by monitoring progress	Project Manager, Developers	All modules completed within the allocated timeline

2.17. PROJECT CHANGE CONTROL

There was an issue with the AI model's detection accuracy. The initial model was not able to effectively identify certain behaviors that were crucial for the system. To address this, we had to make some changes to the detection algorithm and improve the model's performance.

a. We adjusted the algorithm to include additional features for better motion detection, enhancing the model's ability to identify unsafe riding behaviors.

b. We collaborated with external AI specialists to fine-tune the model's accuracy as per the specific requirements of real-time bike riding detection.

2.18. LESSONS LEARNED

Phase	What Worked?	What Did Not Work?	Lessons for Next Project
Initiation Phase	Vendor services, Initial AI model selection	Cleaning and preparation of data	Ensure data quality and preprocessing are prioritized early
Planning Phase	System architecture, Team coordination	Time management, Unrealistic initial timelines	Set more realistic timelines and detailed task allocations
Execution Phase	AI model development, System integration	Filtration issues, Detection model accuracy	Test and optimize the model iteratively, involve specialists earlier